

BANGLADESH UNIVERSITY OF PROFESSIONALS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Digital Communication

Introduction to Communication Channels, Mathematical Models and Characteristics of Communications Channels (Week 1)

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Contents

- Introduction to communication channel
- Characteristics of a Communication Channel
- Mathematical model of a Communication Channel

Introduction to Communication Channel

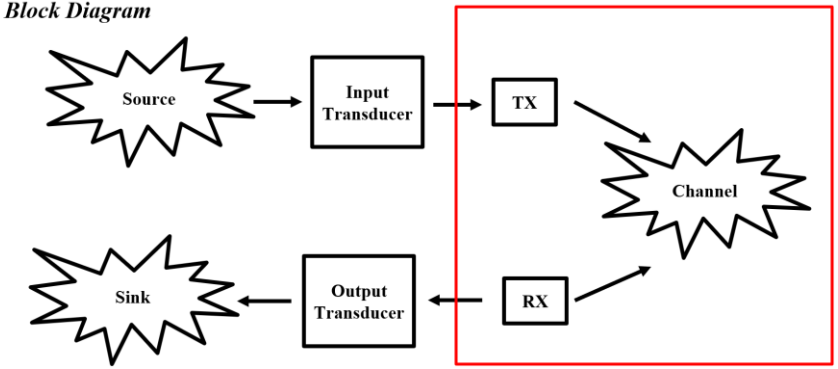
- ***Channel:***

Physical medium used to send signal from transmitter to receiver; have physical properties.

Instead of investigating practically, abstract the channel to mathematical model.

Investigate the characteristics of the channels.

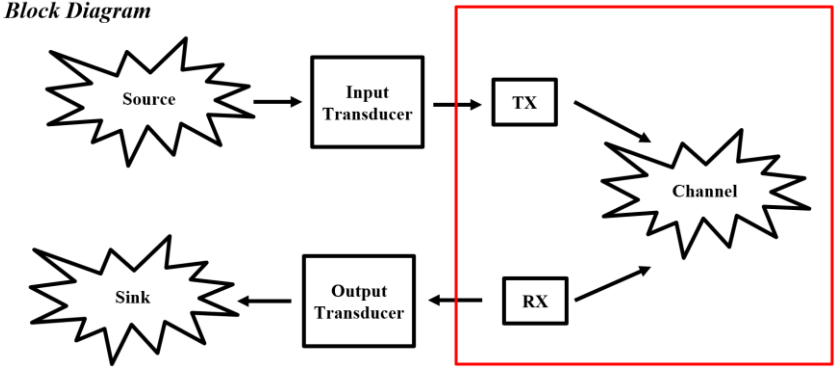
- *Block Diagram*



Introduction to Communication Channel

- ***Channel:***
- Wired
  - Optical (light)
    - Optical fiber.
  - Electrical (voltages and currents)
    - Twisted pair.
    - Co-axial cables.
    - Waveguide.

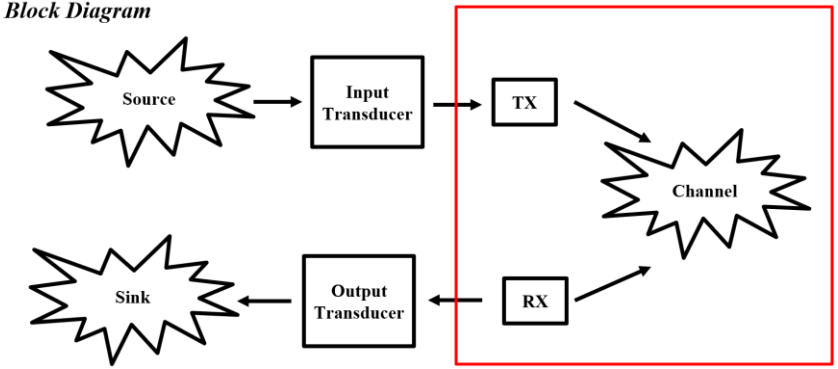
- *Block Diagram*



Introduction to Communication Channel

- ***Channel:***
- Wireless
  - Optical
    - Infrared Channel.
    - Electromagnetic Channel.
      (waves of the optical portion of the spectrum)
  - Radio Waves.
  - Acoustic.
    (Underwater communication- electromagnetic wave do not propagate well under water)

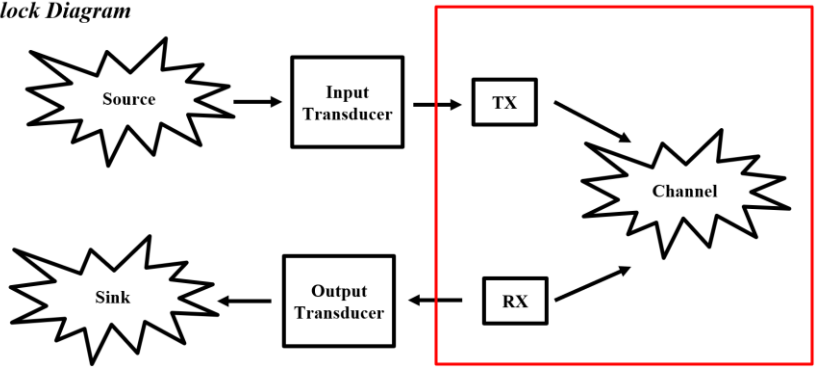
- *Block Diagram*



Introduction to Communication Channel

- ***Channel:***
  - Wired
  - Wireless

- *Block Diagram*



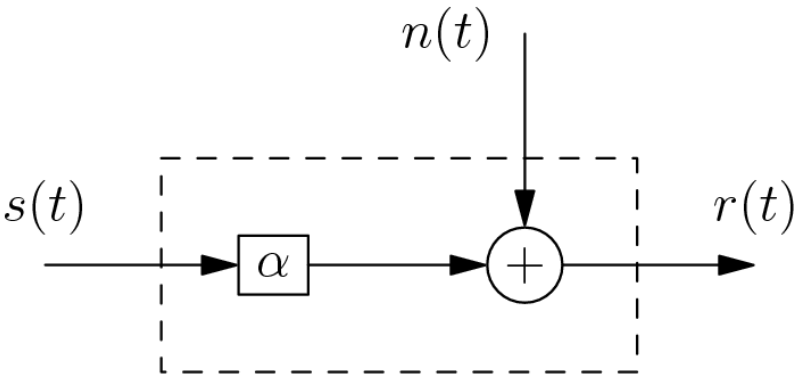
"There is Distortion everywhere"

Characteristics and Mathematical Model of a Comm. Channel

- **Mathematical Models for Channels**

1. Additive Noise Channel
2. Linear Time-Invariant Filter Channel
3. Linear Time-Variant Filter Channel

- **Additive Noise Channel**



Where,

$s(t)$ – transmitted signal

$n(t)$ – additive noise

$r(t)$ – received signal

a – channel gain

Mathematically,

$$r(t) = as(t) + n(t)$$



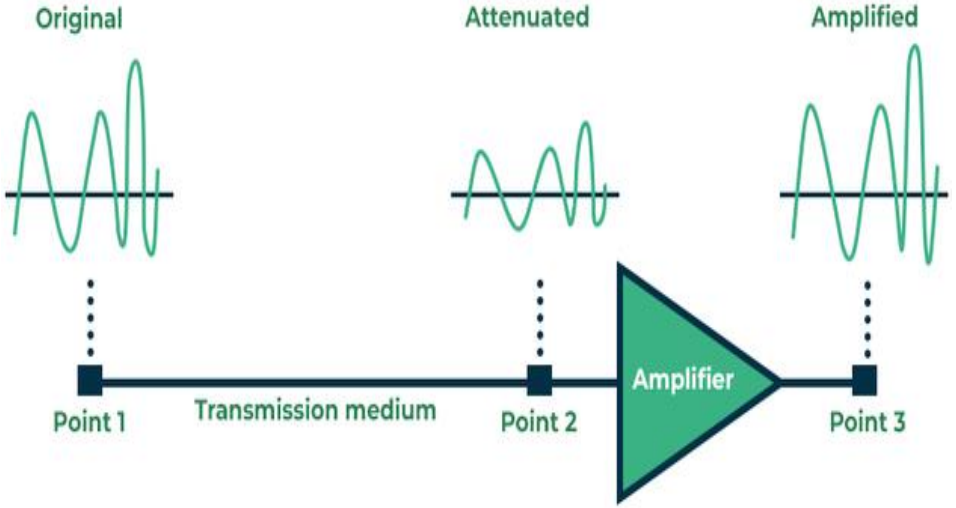


❑ Additive Noise Channel

- Channel Distortions

1. Attenuation –
   - The scaling of the original signal as it passes through a physical medium or channel.
   - If $|a| < 1$,
     It **Attenuates** the signal
   - If $|a| > 1$,
     It **Amplifies** the signal

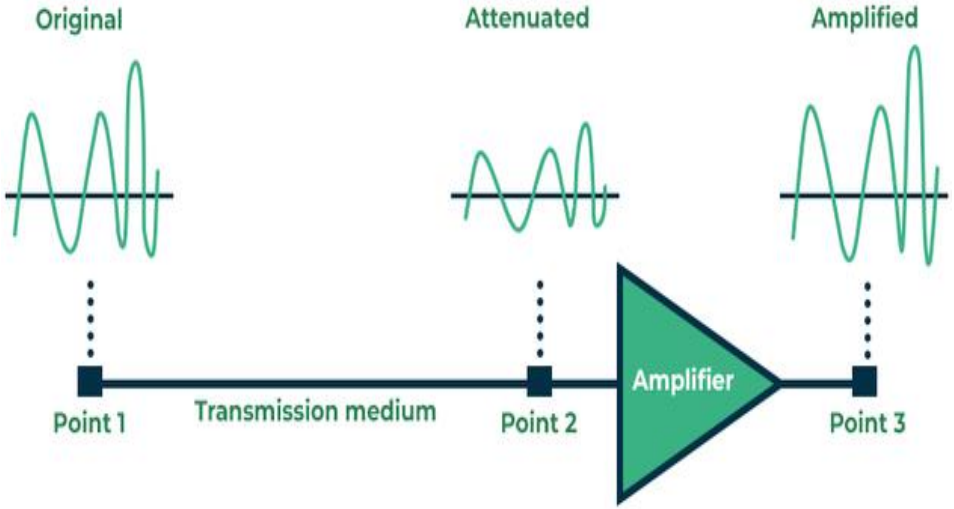
e.g., Active Distributed Antenna Systems (DAS)
Optical Fiber with Optical Amplifiers



❑ Additive Noise Channel

- Channel Distortions

1. Attenuation –
   - $\boldsymbol{a}$ can be negative as well, for 180-degree phase shift (or phase reversal)
   - Can be constant, $\boldsymbol{a}$
   - Can be time varying $\boldsymbol{a}(\boldsymbol{t})$ **Fading Channel** where the signal strength fluctuates over time.



❑ Additive Noise Channel

- Channel Distortions

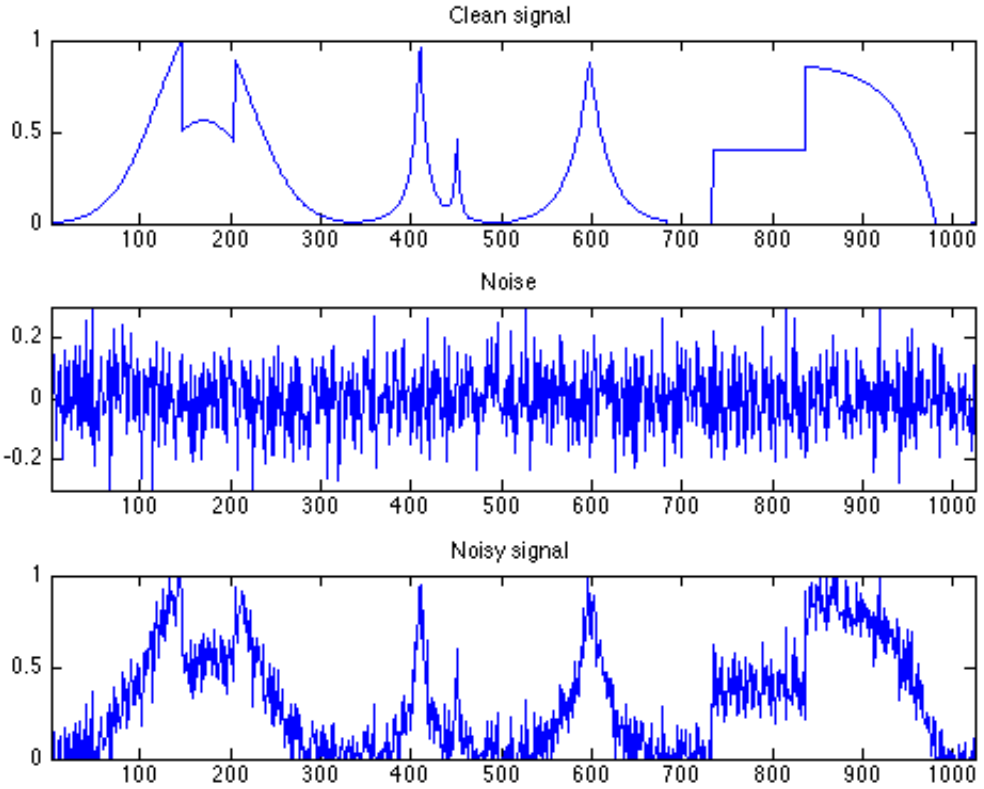
1. Attenuation –
   - a, Critical factor in determining the Signal-to-Noise Ratio (SNR)

$$\text{SNR} = \frac{\text{Power}(a \cdot s(t))}{\text{Power}(n(t))} = \frac{|a|^2 \cdot E[|s(t)|^2]}{E[|n(t)|^2]}$$

Here, for periodic signals

$$E[|s(t)|^2] = \frac{1}{N}\sum_{n=0}^{N-1} |s(t)|^2$$

- **Additive Noise Channel**
  - Channel Distortions
  2. Noise –
     - A type of signal interference that adds directly to an original signal, resulting in a corrupted signal.

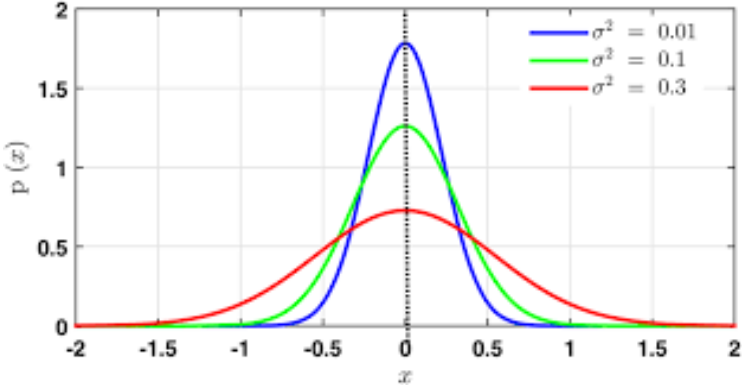


❑ Additive Noise Channel

- Channel Distortions

2. Noise –
    - Gaussian Noise
        1. Statistical Noise: random, unpredictable, unwanted variability in data.
        2. A probability density function (PDF) that follows the Gaussian (normal) distribution.
        3. Small noise values are more probable than large ones.
        1. Often described by its mean (μ) and standard deviation (σ)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Characteristics and Mathematical Model of a Comm. Channel

- **Additive Noise Channel**
  - Channel Distortions
  2. Noise –
     - Gaussian Noise
       1. White Gaussian Noise

          Additive White Gaussian Noise (AWGN).

          Power of each frequency component (read signal) is the same.
       2. Color Gaussian Noise

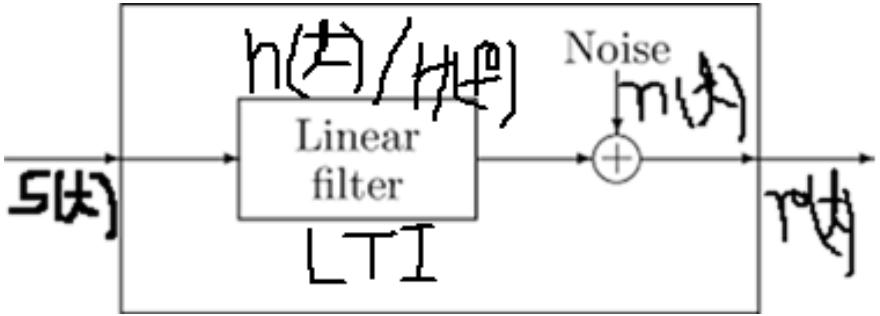
          Additive Color Gaussian Noise (ACGN).

          Power of each frequency component (read signal) is not the same

$$P = \frac{1}{T}\int_0^T |x(t)|^2 dt$$

$$P = \frac{1}{N}\sum_{n=1}^{N} |x[n]|^2$$

- **Linear Time-Invariant (LTI) Filter Channel**



Where,
Linear Filter can be in
Time domain, $h(t)$ (Impulse Response)
Frequency domain, $H(f)$ and

$$h(t) \overset{Fourier\ Transform}{\longleftrightarrow} H(f)$$

Mathematically,

Time Domain: $r(t) = s(t) * h(t) + n(t)$

Frequency Domain: $R(f) = S(f) \cdot H(f) + N(f)$

$h(t)$ *and* $H(f)$ *are kept as constant*

- **Linear Time-Invariant (LTI) Filter Channel**

  Linear Time-Invariant (STI) system

  A system that obeys the principles of

  1. ***Superposition (linearity)***: Output is directly proportional to the input

$$r1(t) \propto s1(t)$$

$$\Rightarrow r1(t) = as1(t)$$

$$r2(t) \propto s1(t)$$

$$\Rightarrow r2(t) = bs2(t)$$

$$r1(s) + r2(s) = as1(t) + bs2(t)$$

- **Linear Time-Invariant (LTI) Filter Channel**

  Linear Time-Invariant (STI) system

  A system that obeys the principles of

  2. ***Time Invariance***

     The system's behavior and characteristics are fixed over time

     $r(t) \propto s(t)$

     $$\Rightarrow r(t) = as(t)$$

     Before or after an elapsed time, τ

     $$\Rightarrow r(t \pm \tau) = as(t \pm \tau)$$

“ মানুষ মরে গেলে পচে যায়, বেঁচে থাকলে বদলায়, কারণে অকারণে বদলায় ”



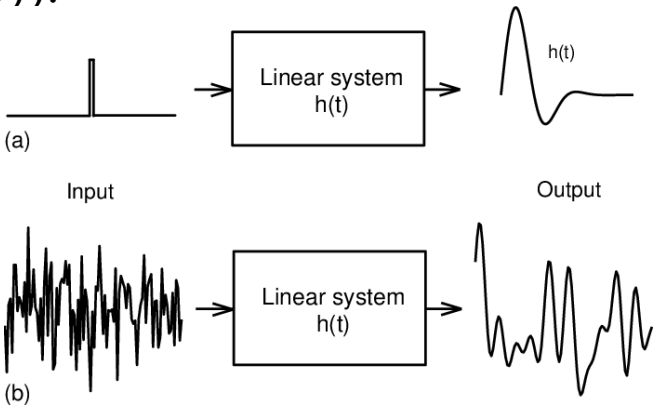
- **Linear Time-Invariant (LTI) Filter Channel**

  Linear Time-Invariant (STI) system

  A system that obeys the principles of

  Impulse Response: The output of the LTI system, $h(t)$ when the input is a unit impulse function ($\delta(t)$).

  A unit impulse signal (Dirac delta function, $\boldsymbol{\delta(t)}$) is a mathematical function that is zero everywhere except a $\boldsymbol{t}$, where it has an infinite amplitude but a total area (integral) of exactly one.



- **Linear Time-Invariant (LTI) Filter Channel**

  Mathematically,

  Time Domain: $r(t) = s(t) * h(t) + n(t)$

$$= \int_{-\infty}^{\infty} s(\tau)h(t-\tau)d\tau + n(t)$$

$$= \int_{-\infty}^{\infty} s(t-\tau)h(\tau)d\tau + n(t)$$

$$\left.\begin{aligned} s(t) * h(t) &= \int_{-\infty}^{\infty} s(\tau)h(t-\tau)d\tau + n(t) \\ &= \int_{-\infty}^{\infty} s(t-\tau)h(\tau)d\tau + n(t) \end{aligned}\right\} \text{Convolution}$$

- **Linear Time-Invariant (LTI) Filter Channel**

  Linear Time-Invariant (STI) system

  A system can also be described with the

  Frequency Response: In an LTI system, For any single frequency ***f***,

  H(f) is a single complex number that acts as the "multiplier" for that specific frequency.

Frequency Domain: $R(f) = \underbrace{S(f) \cdot H(f)}_{\text{Simple Multiplication}} + N(f)$

- **Linear Time-Invariant (LTI) Filter Channel**
  - Channel Distortions
  1. Attenuation
  2. Noise
  3. Modifications of Signal Spectrum
     (Magnitude & Phase)



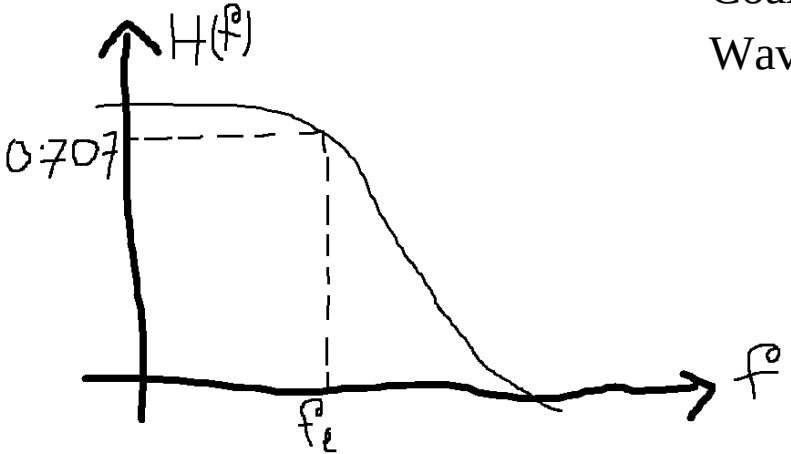
- **Linear Time-Invariant (LTI) Filter Channel**
  - Example

1. Wired Electrical Channel:

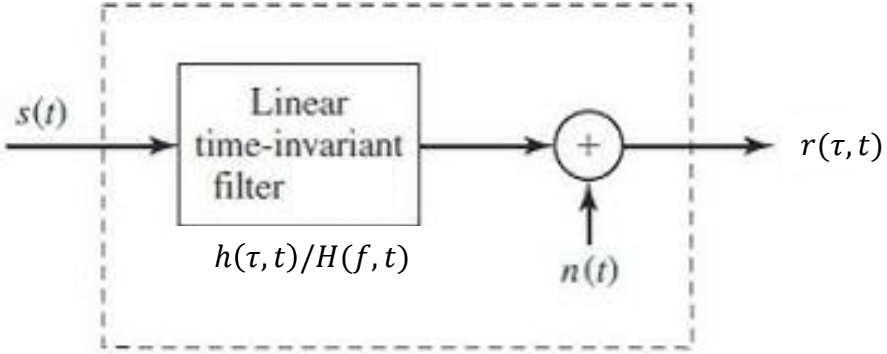
Twisted-pair Cable: $f_c \sim 100kHz$

Coaxial Cable, $f_c \sim 100MHz$

Wave Guide, $f_c \sim 100GHz$



- **Linear Time-Variant (LTV) Filter Channel**



Where,
Linear Filter can be in
Time domain, $h(\tau, t)$ (Impulse Response)
Frequency domain, $H(f, t)$ and

$$h(t) \overset{Fourier\ Transform}{\longleftrightarrow} H(f)$$

Mathematically,

Time Domain $r(\tau, t) = s(t) * h(\tau, t) + n(t)$

$$= \int_{-\infty}^{\infty} s(\mu) h(t - \mu, \tau) d\mu + n(t)$$

Frequency Domain: $R(f, t) = S(f, t) \cdot H(f, t) + N(f)$

❑ Linear Time-Variant (LTV) Filter Channel

- Example

1. Mobile wireless Channel:

   As a user moves, the multipath components (reflections from buildings, cars) change rapidly, making the channel impulse response, $h(t, \tau)$ dependent on the specific time, t.

2. Power-Line Communication (PLC) Channels:

   The channel changes as loads turn on and off, creating time-varying reflections and attenuation.

- **Linear Time-Invariant (LTI) Filter Channel**
  - Channel Distortions
  1. Attenuation
  2. Noise
  3. Modifications of Signal Spectrum
     (Magnitude & Phase)



Summery of the Mathematical Models

1. Additive Noise Channel:

$$r(t) = as(t) + \boldsymbol{n(t)}$$

2. Linear Time-Invariant Filter Channel:

Time Domain: $r(t) = s(t) * h(t) + \boldsymbol{n(t)}$

Frequency Domain: $R(f) = S(f) \cdot H(f) + \boldsymbol{N(f)}$

3. Linear Time-Variant Filter Channel:

Time Domain $r(\tau, t) = s(t) * h(\tau, t) + \boldsymbol{n(t)}$

Frequency Domain: $R(f, t) = S(f, t) \cdot H(f, t) + \boldsymbol{N(f)}$

Gratitude

Thank you very much

Jaja Kallah Khairan